STK4021 – Mandatory assignment, Fall 2026

Bayesian inversion of seismic data

NAME – STK4021

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Self-report

To be written by the student: how the work proceeded, what was difficult, time-consuming or fun, what was relatively easy, collaboration and sources used (including AI tools).

Data and forward operator

The forward operator $\mathbf{A}$ is a 101×101 matrix. Each row contains the same wavelet, shifted one sample, applied to differences of $\mathbf{m}$. In particular $\mathbf{A1} = \mathbf{0}$, so a constant shift of the profile is invisible in the data.

Part 1: Gaussian Bayesian inversion and simulation

1a) Joint distribution and posterior

Let $\sim N(\boldsymbol{\mu}_m, \boldsymbol{\Sigma}_m)$ and $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I})$ be independent, and $\mathbf{d} = \mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}$. The vector $(\mathbf{m}, \mathbf{d})$ is a linear transformation of the Gaussian vector $(\mathbf{m}, \boldsymbol{\varepsilon})$, so it is jointly Gaussian, and it is enough to find the first two moments:

$$\begin{aligned} E(\mathbf{d}) &= \mathbf{A}E(\mathbf{m}) + E(\boldsymbol{\varepsilon}) = \mathbf{A}\boldsymbol{\mu}_m, \\ \mathrm{Cov}(\mathbf{d}) &= \mathbf{A}\,\mathrm{Cov}(\mathbf{m})\mathbf{A}^\top + \mathrm{Cov}(\boldsymbol{\varepsilon}) = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I}, \\ \mathrm{Cov}(\mathbf{d}, \mathbf{m}) &= \mathrm{Cov}(\mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}, \mathbf{m}) = \mathbf{A}\boldsymbol{\Sigma}_m, \end{aligned}$$

using independence of $\mathbf{m}$ and $\boldsymbol{\varepsilon}$. Hence

$$\begin{pmatrix} \mathbf{m} \\ \mathbf{d} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_m \\ \mathbf{A}\boldsymbol{\mu}_m \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_m & \boldsymbol{\Sigma}_m\mathbf{A}^\top \\ \mathbf{A}\boldsymbol{\Sigma}_m & \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I} \end{pmatrix} \right).$$

By the conditioning rule for the multivariate Gaussian, the posterior is $\mathbf{m} \mid \mathbf{d} \sim N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$ with

$$\begin{aligned} \boldsymbol{\mu}_{m|d} &= \boldsymbol{\mu}_m + \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}(\mathbf{d} - \mathbf{A}\boldsymbol{\mu}_m), \\ \boldsymbol{\Sigma}_{m|d} &= \boldsymbol{\Sigma}_m - \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}\mathbf{A}\boldsymbol{\Sigma}_m, \qquad \boldsymbol{\Sigma}_d = \mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I}. \end{aligned}$$

The matrix $\mathbf{K} = \boldsymbol{\Sigma}_m\mathbf{A}^\top\boldsymbol{\Sigma}_d^{-1}$ is the "gain": the posterior mean is the prior mean corrected by $\mathbf{K}$ times the data residual, and the posterior covariance does not depend on the observed $\mathbf{d}$.

1b) Prior: mean, 95% interval and simulations

With $(\mu, \sigma_m^2, \ell, \sigma_\varepsilon^2) = (0, 1, 20, 0.04)$ the prior is stationary, so the marginal $m_i \sim N(0, 1)$ for every i and the 95% interval is 0 ± 1.96. Five realisations are drawn as $\mathbf{m} = \boldsymbol{\mu}_m + \mathbf{L}^\top\mathbf{z}$ with $\boldsymbol{\Sigma}_m = \mathbf{L}^\top\mathbf{L}$ (Cholesky) and $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$. The correlation $\exp(-3|i-j|/\ell)$ drops to $e^{-3} \approx 0.05$ at lag $\ell = 20$, so the realisations are rough (exponential correlation) but vary on a scale of about 20 samples.

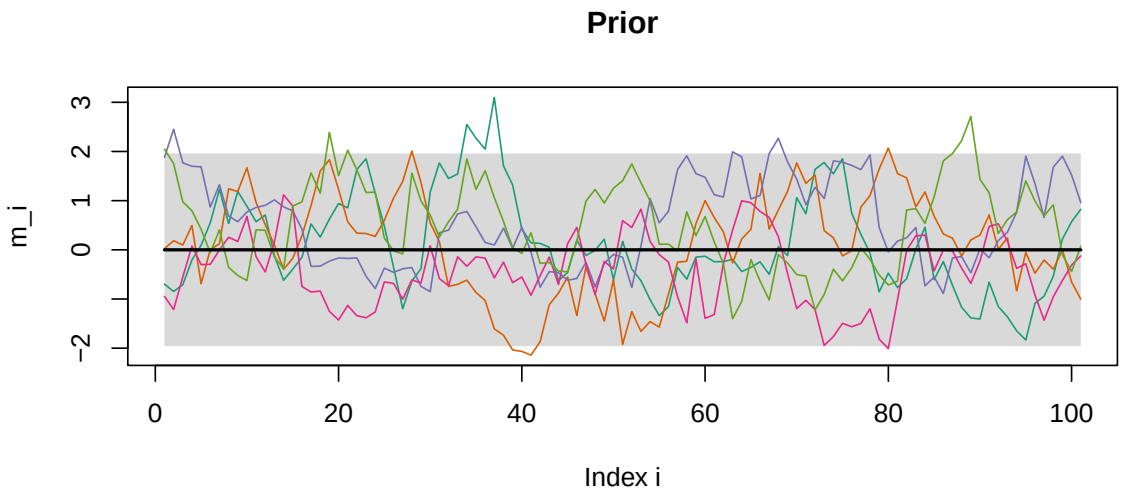


Figure 1: Prior mean (black), pointwise 95% interval (grey) and five prior realisations.

1c) Posterior given `seis00`

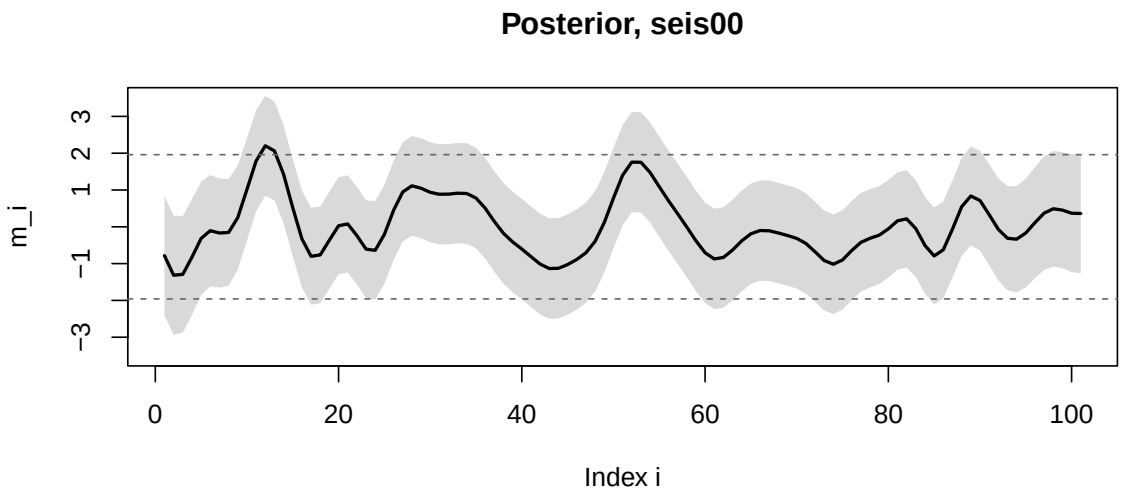


Figure 2: Posterior mean (black) and pointwise 95% interval (grey) for seis00. Dashed: prior 95% limits.

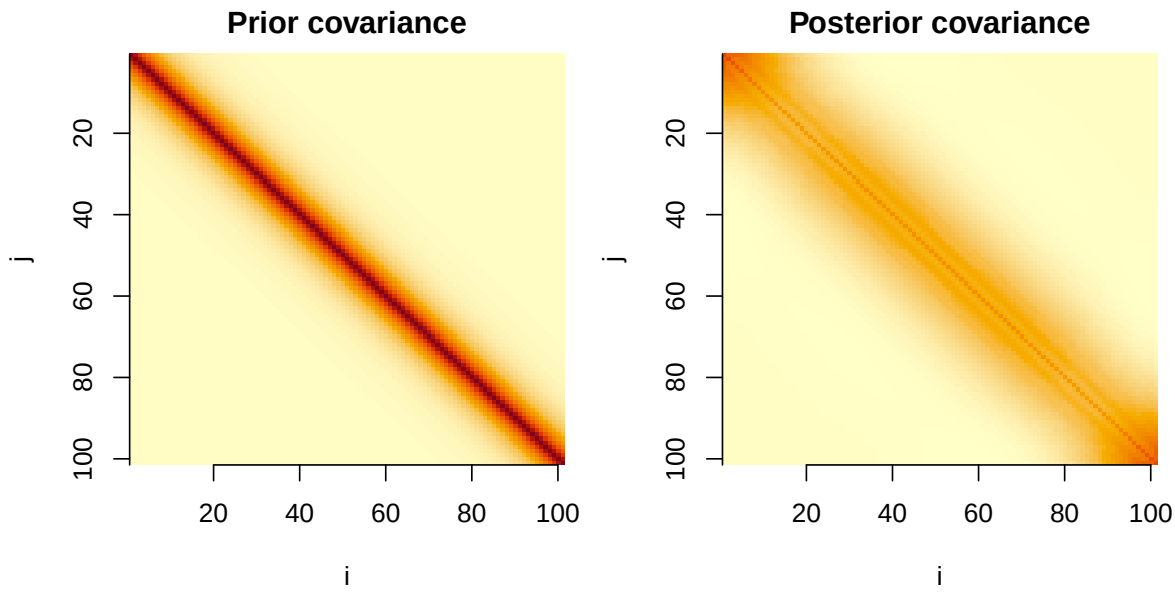


Figure 3: Prior (left) and posterior (right) covariance matrices, same colour scale.

The posterior standard deviations lie in [0.67, 0.83] compared with 1 in the prior. They are smallest in the interior and largest at the two ends, where fewer data rows see the parameters. The covariance images show what the data have resolved. The posterior variances are roughly halved, but the remaining covariance is spread much more widely around the diagonal, so posterior correlations decay slowly with lag. The wavelet in $\mathbf{A}$ is band limited, and since $\mathbf{A1} = \mathbf{0}$ it cannot see the overall level or slow trends. The data therefore determine the short-scale *contrasts* in $\mathbf{m}$, while the long-wavelength part of the profile keeps almost all of its prior uncertainty. For example, the variance of the profile average $\bar{m} = \frac{1}{n}\sum_i m_i$ is 0.124 a priori and 0.121 a posteriori.

1d) Exceedance volume

$V = h(\mathbf{m}) = \sum_i |m_i - c|\, I(m_i < c)$ with $c = -1$ is a non-linear function of $\mathbf{m}$, so its posterior is found by Monte Carlo. We draw $\mathbf{m}^{(1)}, \ldots, \mathbf{m}^{(B)}$ iid from $N(\boldsymbol{\mu}_{m|d}, \boldsymbol{\Sigma}_{m|d})$ and set $V^{(b)} = h(\mathbf{m}^{(b)})$.

Table 1: Posterior summaries of V from $B = 100$ samples.

mean	median	sd
7.94	5.14	7.56

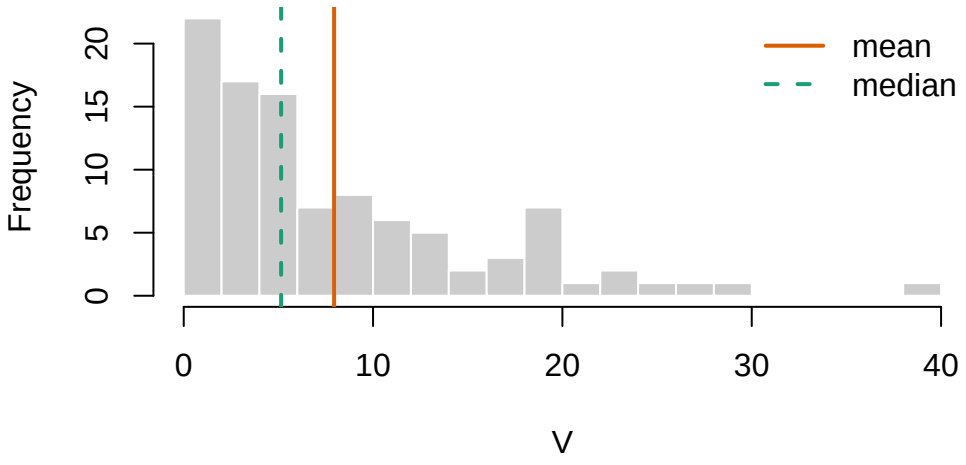


Figure 4: Histogram of 100 posterior samples of V.

The distribution is strongly right-skewed: most profiles dip only slightly below c, while a few have long or deep excursions. Hence the mean is clearly larger than the median, and the standard deviation is of the same size as the mean.

1e) Percentiles and their uncertainty

A natural estimate of P_{10} is the 10th smallest sample, and of P_{90} the 90th: $\widehat{P}_{10} = V_{(10)} = 1.02$ and $\widehat{P}_{90} = V_{(90)} = 18.88$.

For the uncertainty, let $N = \#\{b : V^{(b)} < P_{10}\}$. Each sample falls below P_{10} with probability 0.1, independently, so $N \sim \mathrm{Bin}(B, 0.1)$. The event $V_{(r)} \le P_{10} < V_{(s)}$ is the same as $r \le N \le s-1$, so

$$P\big(V_{(r)} \le P_{10} < V_{(s)}\big) = \sum_{k=r}^{s-1} \binom{B}{k} 0.1^k 0.9^{B-k}.$$

With $r = 5$ and $s = 17$ this probability is 0.956, which gives the distribution-free interval $[V_{(5)}, V_{(17)}] = [0.4,\ 1.5]$ for P_{10}. The interval is wide relative to $\widehat{P}_{10}$, so $B = 100$ is too small for a precise P_{10}. The relative width shrinks like $1/\sqrt{B}$. For example, with $B = 10\,000$ the same construction uses order statistics around 1000 ± 59.

1f) Permutation test of exchangeability

If the data were exchangeable, every ordering of $\mathbf{d}$ would be equally likely, and the lag-1 correlation $\hat{\rho}_1 = \mathrm{cor}(d_{1:n-1}, d_{2:n})$ of the observed ordering would look like a typical value from random permutations.

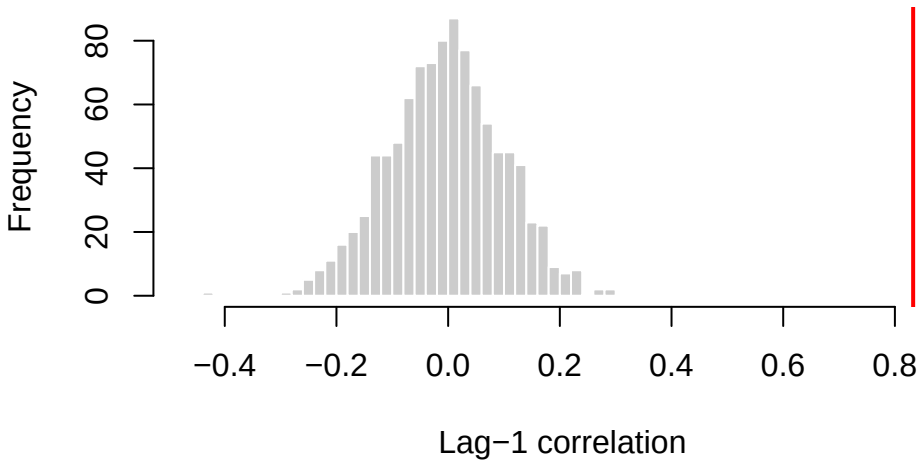


Figure 5: Lag-1 correlation for 1000 random permutations of seis00 (histogram) and for the observed ordering (red line).

The observed lag-1 correlation is $\hat{\rho}_1 = 0.833$, while the permuted values lie in [-0.42, 0.3] (centred near $-1/n$). None of the 1000 permutations come close, so the Monte Carlo p-value is 0.001 (the smallest possible with 1000 permutations), and exchangeability is clearly rejected. This is what the model predicts: under $\mathbf{d} = \mathbf{A}\mathbf{m} + \boldsymbol{\varepsilon}$ the covariance $\mathbf{A}\boldsymbol{\Sigma}_m\mathbf{A}^\top + \sigma_\varepsilon^2\mathbf{I}$ is far from the form $a\mathbf{I} + b\mathbf{1}\mathbf{1}^\top$ that exchangeability requires. Neighbouring samples share the wavelet and are strongly correlated. An iid (or exchangeable) model for the seismic trace is therefore not appropriate, while the convolution model is consistent with the dependence we see.

Appendix: R code

```
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE,
                      fig.align = "center", fig.width = 7.5, fig.height = 3.6,
                      out.width = "95%", fig.pos = "H")
set.seed(4021)
A  <- as.matrix(read.table("data/A.txt"))
dimnames(A) <- NULL
d0 <- scan("data/seis00.txt", quiet = TRUE)
d1 <- scan("data/seis01.txt", quiet = TRUE)
d2 <- scan("data/seis02.txt", quiet = TRUE)
n  <- nrow(A)
z  <- 1:n   # index (depth/time sample)
# Prior covariance with exponential correlation function
exp_cov <- function(n, sigma2, ell) {
  D <- abs(outer(1:n, 1:n, "-"))
  sigma2 * exp(-3 * D / ell)
}

# Gaussian posterior for m | d in the linear model d = A m + eps
gauss_posterior <- function(d, A, mu, Sigma, sigma2_eps) {
  Sigma_d <- A %*% Sigma %*% t(A) + sigma2_eps * diag(nrow(A))
  K       <- Sigma %*% t(A) %*% solve(Sigma_d)        # gain matrix
  mu_post <- drop(mu + K %*% (d - A %*% mu))
  S_post  <- Sigma - K %*% A %*% Sigma
  S_post  <- (S_post + t(S_post)) / 2                 # remove rounding asymmetry
  list(mean = mu_post, cov = S_post, K = K)
}

# Draw B samples from N(mu, S); rows are samples
rmvn <- function(B, mu, S) {
  L <- chol(S + 1e-10 * diag(length(mu)))             # tiny jitter for stability
  sweep(matrix(rnorm(B * length(mu)), B) %*% L, 2, mu, "+")
}

# Plot mean with pointwise 95% interval
band_plot <- function(mean, sd, main, ylim = NULL, sims = NULL) {
  lo <- mean - qnorm(0.975) * sd
  hi <- mean + qnorm(0.975) * sd
  if (is.null(ylim)) ylim <- range(lo, hi, sims)
  plot(z, mean, type = "n", ylim = ylim, xlab = "Index i", ylab = "m_i", main = main)
  polygon(c(z, rev(z)), c(lo, rev(hi)), col = "grey85", border = NA)
  if (!is.null(sims)) matlines(z, t(sims), lty = 1, lwd = 1,
                               col = c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e"))
  lines(z, mean, lwd = 2)
}
mu0 <- 0; sigma2_m <- 1; ell <- 20; sigma2_eps <- 0.04
mu_m    <- rep(mu0, n)
Sigma_m <- exp_cov(n, sigma2_m, ell)

prior_sims <- rmvn(5, mu_m, Sigma_m)
band_plot(mu_m, sqrt(diag(Sigma_m)), "Prior", sims = prior_sims)
post0 <- gauss_posterior(d0, A, mu_m, Sigma_m, sigma2_eps)
```

```r
sd0   <- sqrt(diag(post0$cov))
band_plot(post0$mean, sd0, "Posterior, seis00", ylim = c(-3.5, 3.5))
abline(h = c(-1, 1) * 1.96, lty = 2, col = "grey40")
cov_image <- function(S, main, zlim) {
  image(z, z, S[, n:1], zlim = zlim, col = hcl.colors(64, "YlOrRd", rev = TRUE),
        xlab = "i", ylab = "j", main = main, axes = FALSE, useRaster = TRUE)
  axis(1); axis(2, at = n + 1 - pretty(z), labels = pretty(z))
}
zl <- range(Sigma_m, post0$cov)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))
cov_image(Sigma_m, "Prior covariance", zl)
cov_image(post0$cov, "Posterior covariance", zl)
sd_range <- range(sd0)
w <- rep(1 / n, n)                                  # weights for the profile average
var_level <- c(prior = drop(t(w) %*% Sigma_m %*% w),
               posterior = drop(t(w) %*% post0$cov %*% w))
c_lev <- -1
h_V   <- function(m, c = c_lev) sum((c - m) * (m < c))

B <- 100
post_sims <- rmvn(B, post0$mean, post0$cov)
V <- apply(post_sims, 1, h_V)

V_summary <- c(mean = mean(V), median = median(V), sd = sd(V))
knitr::kable(t(round(V_summary, 2)), caption = "Posterior summaries of $V$ from $B=100$ samples.")
hist(V, breaks = 20, col = "grey80", border = "white", main = "", xlab = "V")
abline(v = V_summary["mean"], lty = 1, lwd = 2, col = "#d95f02")
abline(v = V_summary["median"], lty = 2, lwd = 2, col = "#1b9e77")
legend("topright", c("mean", "median"), lty = 1:2, lwd = 2,
       col = c("#d95f02", "#1b9e77"), bty = "n")
Vs  <- sort(V)
P10 <- Vs[10]; P90 <- Vs[90]

# Number of samples below the true P10 is N ~ Bin(B, 0.1).
# The interval [V_(r), V_(s)) contains P10 exactly when r <= N <= s - 1.
r <- qbinom(0.025, B, 0.1)            # lower order statistic
s <- qbinom(0.975, B, 0.1) + 1        # upper order statistic
coverage <- pbinom(s - 1, B, 0.1) - pbinom(r - 1, B, 0.1)
lag1 <- function(x) cor(x[-1], x[-length(x)])
rho_obs  <- lag1(d0)
rho_perm <- replicate(1000, lag1(sample(d0)))
p_value  <- (1 + sum(abs(rho_perm) >= abs(rho_obs))) / (1 + length(rho_perm))

hist(rho_perm, breaks = 30, col = "grey80", border = "white", main = "",
     xlab = "Lag-1 correlation", xlim = range(rho_perm, rho_obs) + c(-0.05, 0.05))
abline(v = rho_obs, col = "red", lwd = 2)
```